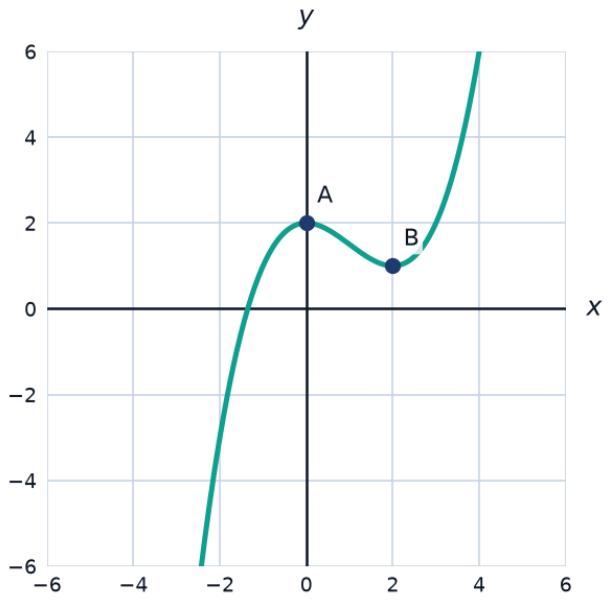
Polynomial End Behavior & Zeros

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Leading Term Test and End Behavior**

A polynomial $f(x)$ of degree 4 with real coefficients has leading coefficient 2 and zeros at $x = -1$ (multiplicity 2) and $x = 2 + i$. Write $f(x)$ in fully expanded standard form.



2 **[Foundational]** • **Target: Multiplicity of Zeros and Graph Geometry**

When the polynomial $P(x) = 2x^3 + kx^2 - 7x + 6$ is divided by $(x - 2)$, the remainder is 12. Find the value of k, and determine whether $(x + 3)$ is a factor of $P(x)$.

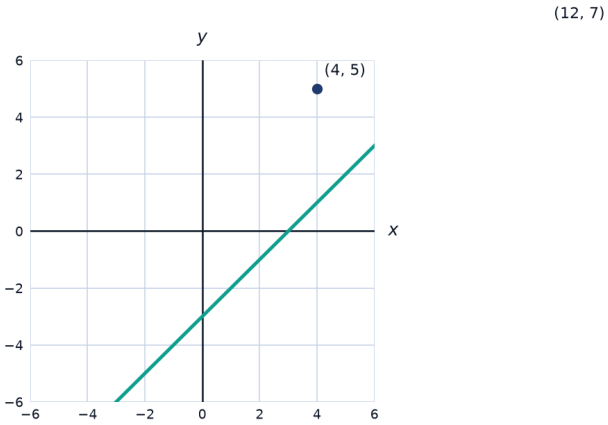
3 **[Foundational]** • **Target: Remainder & Factor Theorems**

Determine the end behavior of $P(x) = -4x^5 + 7x^3 - 2x + 9$ as $x \to -\infty$ and as $x \to +\infty$ using limit notation, and state the maximum number of local extrema.

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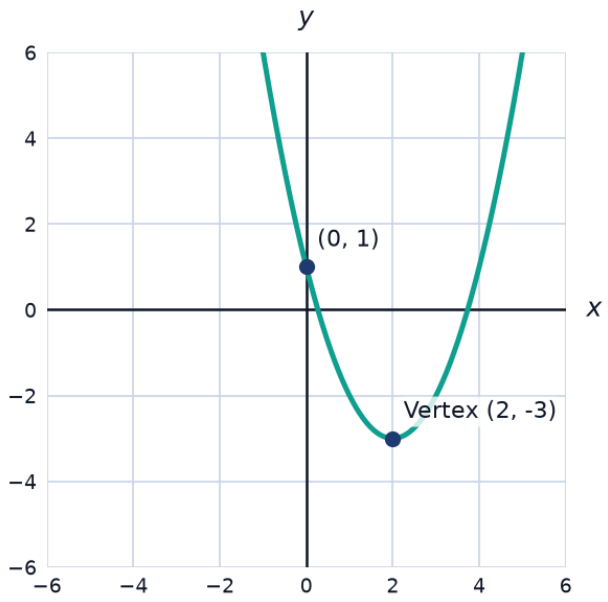
4 [Foundational] • Target: Rational Root Theorem & Complex Conjugates

The graph of the exponential function $f(x) = a \cdot \log_3(x - h) + k$ is shown below. The function has a vertical asymptote at $x = 3$ and passes through the points $(4, 5)$ and $(12, 7)$. Determine the value of the parameter a.



5 [Foundational] • Target: Leading Term Test and End Behavior

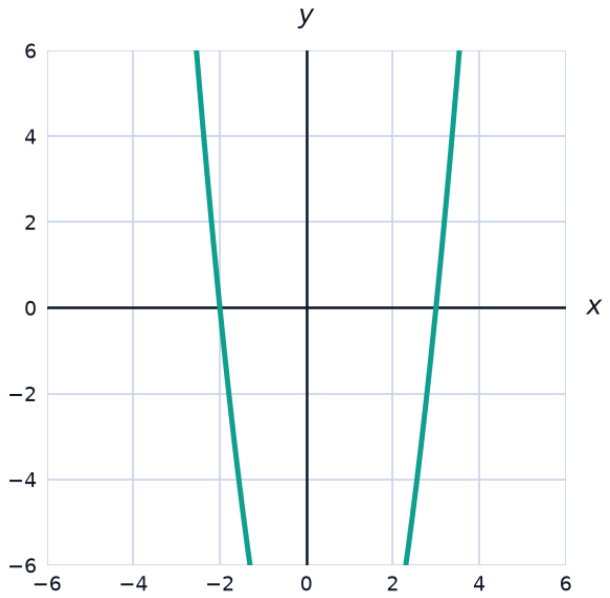
The graph of the quadratic function $f(x) = x^2 - bx + c$ is shown below. The vertex of the parabola is at $(2, -3)$, and it passes through the point $(0, 1)$. Determine the value of the parameter b.



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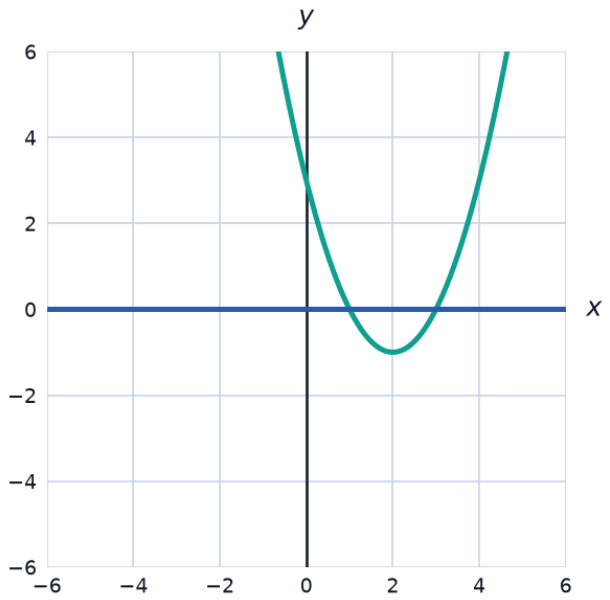
6 [Foundational] • Target: Multiplicity of Zeros and Graph Geometry

For the rational function $r(x) = \frac{2x^2 - 2x - 12}{x^2 - 9}$, find: (a) the domain, (b) the coordinates of any removable discontinuity (hole), (c) the vertical and horizontal asymptotes, and (d) the x- and y-intercepts.



7 [Foundational] • Target: Remainder & Factor Theorems

Consider the quadratic function $f(x) = x^2 - 4x + 3$. What are the x-intercepts of the graph of this function?

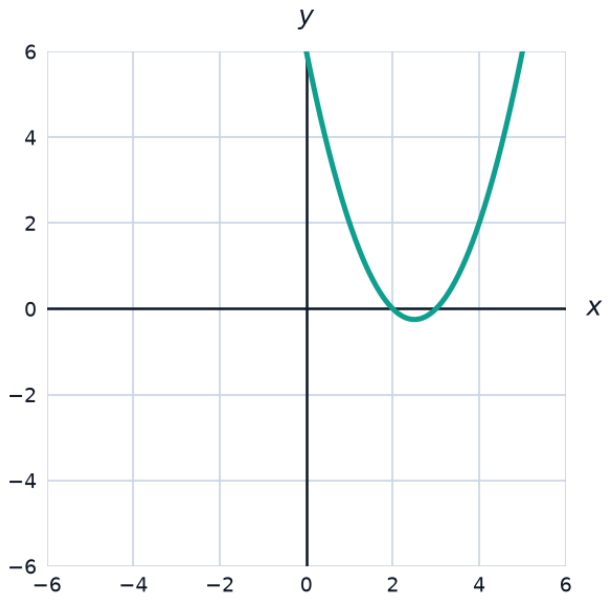


Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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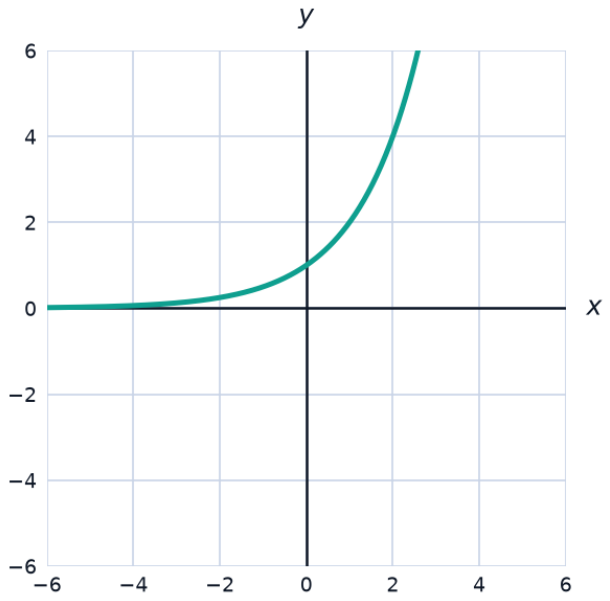
8 **[Intermediate]** • **Target: Rational Root Theorem & Complex Conjugates**

Consider the polynomial function $f(x) = x^2 - 5x + 6$. What are the roots of the function $f(x)$?



9 **[Intermediate]** • **Target: Leading Term Test and End Behavior**

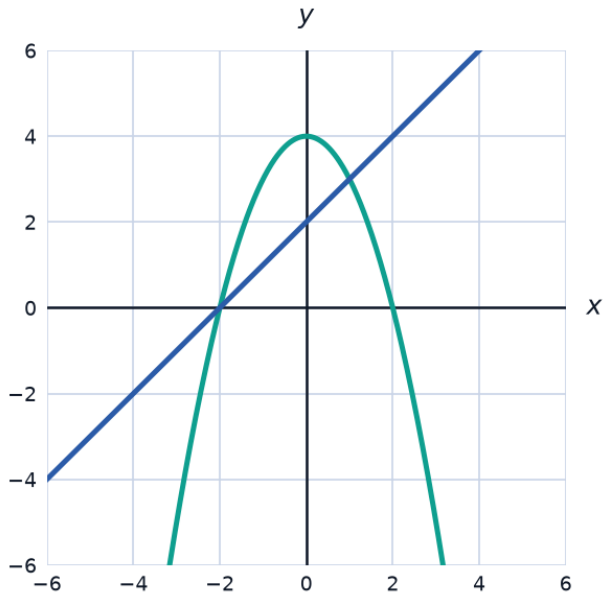
The graph of the exponential function $f(x) = 2^x$ is given. A point P on the graph has a y-coordinate of 8. What is the x-coordinate of point P?



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10 **[Intermediate]** • **Target: Multiplicity of Zeros and Graph Geometry**

Find the average rate of change of the polynomial function $f(x) = 2x^3 - 5x + 4$ over the closed interval $[-1, 2]$.



11 **[Intermediate]** • **Target: Remainder & Factor Theorems**

Solve the rational inequality $\frac{x+4}{x-2} \leq 3$ and express the solution set in interval notation.

12 **[Intermediate]** • **Target: Rational Root Theorem & Complex Conjugates**

Find the equation of the slant (oblique) asymptote of the rational function $g(x) = \frac{3x^2 - 5x + 7}{x - 2}$, and compute $\lim_{x \to 2^+} g(x)$.

13 **[Intermediate]** • **Target: Leading Term Test and End Behavior**

The graph of $y = f(x)$ is transformed into $y = -2f(3x - 6) + 5$. Describe the exact sequence of geometric transformations applied to $f(x)$, and find the image of the point $(3, -1)$ on $y = f(x)$.

14 **[Intermediate]** • **Target: Multiplicity of Zeros and Graph Geometry**

Let $z_1 = 2\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$ and $z_2 = 3\left(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12}\right)$. Find the product $z_1 z_2$ in rectangular form $a + bi$, and compute z_1^6 using De Moivre's Theorem.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Remainder & Factor Theorems** • **Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs**

Solve the polynomial inequality $x^3 - 4x^2 + x + 6 \leq 0$.

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16 [Challenging] • Target: Rational Root Theorem & Complex Conjugates • Cross-Lesson: 4.01 Inverse & Composite Functions

A geometric sequence has third term $g_3 = 18$ and sixth term $g_6 = -\frac{16}{3}$. Find the common ratio r, the first term g_1, and the infinite sum S_∞.

17 [Challenging] • Target: Leading Term Test and End Behavior • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

A log-log regression plot of $(\ln x, \ln y)$ produces the linear fit $\ln(\hat{y}) = \ln(5) - 2.5\ln(x)$ for $x > 0$. Write $\hat{y}$ as an explicit power function of x, and find the factor by which $\hat{y}$ changes when x is multiplied by 4.

18 [Challenging] • Target: Multiplicity of Zeros and Graph Geometry • Cross-Lesson: 4.03 Parametric Equations & Modeling

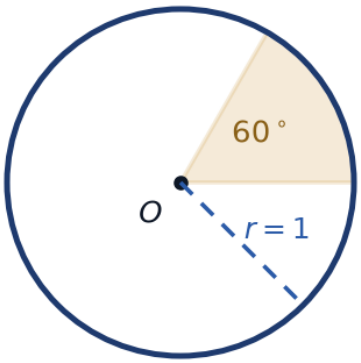
Use the Binomial Theorem to find the coefficient of x^3y^2 in the expansion of $(2x - 3y)^5$.

19 [Challenging] • Target: Remainder & Factor Theorems • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Find the 12th term and the sum of the first 25 terms of the arithmetic sequence with $a_1 = -7$ and common difference $d = 4$.

20 [Challenging] • Target: Rational Root Theorem & Complex Conjugates • Cross-Lesson: 1.03 Rates of Change & Concavity

A point P is on the unit circle such that its terminal ray makes an angle of θ with the positive x-axis. The coordinates of P are $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$. If a point Q on the unit circle has an angle ϕ such that $\phi = \theta + \frac{\pi}{3}$, what is the y-coordinate of point Q?



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Leading Term Test and End Behavior

Determine the end behavior of the polynomial function $f(x) = -3x^5 + 7x^3 - 2x + 9$.

a. As $x \to \infty$, $f(x) \to -\infty$; as $x \to -\infty$, $f(x) \to \infty$

b. As $x \to \infty$, $f(x) \to \infty$; as $x \to -\infty$, $f(x) \to -\infty$

c. As $x \to \infty$, $f(x) \to -\infty$; as $x \to -\infty$, $f(x) \to -\infty$

d. As $x \to \infty$, $f(x) \to \infty$; as $x \to -\infty$, $f(x) \to \infty$

22 [Foundational] • Target: Leading Term Test and End Behavior

If a polynomial $p(x)$ satisfies $\lim_{x \to \infty} p(x) = -\infty$ and $\lim_{x \to -\infty} p(x) = -\infty$, which of the following could be the leading term of $p(x)$?

a. $-4x^6$

b. $2x^6$

c. $-5x^7$

d. $3x^5$

23 [Foundational] • Target: Leading Term Test and End Behavior

Consider $g(x) = (2 - x)(x + 3)^2(2x - 1)$. What is the leading coefficient of $g(x)$?

a. -2

b. 2

c. -1

d. 4

24 [Foundational] • Target: Multiplicity of Zeros and Graph Geometry

Let $f(x) = (x - 4)^3(x + 2)^2(x - 1)$. At which zero does the graph of f touch the x-axis and turn around without crossing?

a. $x = -2$

b. $x = 4$

c. $x = 1$

d. $x = 2$

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25 [Foundational] • Target: Multiplicity of Zeros and Graph Geometry

A polynomial $p(x)$ of degree 4 has a zero at $x = 3$ of multiplicity 2, and no other real zeros. If $p(0) = 18$, which equation represents $p(x)$ if its remaining zeros are non-real complex conjugates $\pm i$?

a. $p(x) = 2(x-3)^2(x^2+1)$

b. $p(x) = (x-3)^2(x^2+1)$

c. $p(x) = 2(x-3)^2(x^2-1)$

d. $p(x) = \frac{1}{2}(x-3)^2(x^2+1)$

26 [Foundational] • Target: Multiplicity of Zeros and Graph Geometry

At $x = c$, $f(x) = (x-c)^3 g(x)$ where $g(c) \neq 0$. What is the behavior of the graph of f at $(c, 0)$?

a. The graph crosses the x-axis with an inflection/flattening point (horizontal tangent)

b. The graph touches the x-axis and turns around symmetrically

c. The graph has a vertical asymptote at $x = c$

d. The graph crosses the x-axis linearly without flattening

27 [Foundational] • Target: Remainder & Factor Theorems

When $f(x) = 2x^3 - 5x^2 + kx - 8$ is divided by $(x-3)$, the remainder is 7. Find the value of k.

a. -4

b. 4

c. -2

d. 5

28 [Foundational] • Target: Remainder & Factor Theorems

Which of the following is a factor of $P(x) = x^3 - 6x^2 + 11x - 6$?

a. $x - 2$

b. $x + 2$

c. $x - 4$

d. $x + 1$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Remainder & Factor Theorems

A polynomial $q(x)$ satisfies $q(-4) = 0$, $q(1) = 0$, and $q(5) = 12$. Which statement must be true?

a. $(x + 4)$ and $(x - 1)$ are factors of $q(x)$

b. $(x - 4)$ and $(x + 1)$ are factors of $q(x)$

c. The remainder when $q(x)$ is divided by $(x + 5)$ is 12

d. $q(x)$ must be a quadratic polynomial

30 [Intermediate] • Target: Rational Root Theorem & Complex Conjugates

According to the Rational Root Theorem, which of the following is NOT a possible rational zero of $f(x) = 3x^3 - 2x^2 + 5x - 4$?

a. $\frac{3}{2}$

b. $\frac{4}{3}$

c. ± 1

d. ± 2

31 [Intermediate] • Target: Rational Root Theorem & Complex Conjugates

A polynomial with real coefficients has degree 3. Two of its zeros are 4 and $2 - 3i$. What is the third zero?

a. $2 + 3i$

b. $-2 + 3i$

c. -4

d. $-2 - 3i$

32 [Intermediate] • Target: Rational Root Theorem & Complex Conjugates

What is the total number of complex zeros (counting multiplicity) of $P(x) = (x^2 + 4)^2(x - 5)^3$?

a. 7

b. 5

c. 4

d. 3

33 [Intermediate] • Target: Leading Term Test and End Behavior

Consider a cubic polynomial $P(x) = ax^3 + bx^2 + cx + d$. The graph of $P(x)$ has a y-intercept at $(0, 12)$. It is known that $P(x)$ has a local maximum at $x = -1$ and a local minimum at $x = 3$. If the value of the local maximum is 20, determine the value of the constant term d of the polynomial.

a. 12

b. 10

c. 8

d. 6

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34 [Intermediate] • Target: Multiplicity of Zeros and Graph Geometry

For the polynomial $f(x) = -1x^3 + 3x - 1$, describe the end behavior of f.

a. down on the left, up on the right (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

b. down on both ends (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

c. up on the left, down on the right (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

d. up on both ends (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

35 [Intermediate] • Target: Remainder & Factor Theorems

For the polynomial $f(x) = -2x^3 + 3x - 1$, describe the end behavior of f.

a. up on the left, down on the right (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

b. down on both ends (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

c. up on both ends (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

d. down on the left, up on the right (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

36 [Intermediate] • Target: Rational Root Theorem & Complex Conjugates

For the polynomial $f(x) = -3x^5 + 3x - 1$, describe the end behavior of f.

a. up on both ends (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

b. up on the left, down on the right (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

c. down on the left, up on the right (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

d. down on both ends (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

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37 [Intermediate] • Target: Leading Term Test and End Behavior

For the polynomial $f(x) = -1x^5 + 3x - 1$, describe the end behavior of f.

a. up on the left, down on the right (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

b. up on both ends (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

c. down on the left, up on the right (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

d. down on both ends (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiplicity of Zeros and Graph Geometry • Cross-Lesson: 4.03 Parametric Equations & Modeling

For the polynomial $f(x) = -2x^5 + 3x - 1$, describe the end behavior of f.

a. up on the left, down on the right (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

b. down on both ends (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

c. up on both ends (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

d. down on the left, up on the right (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

39 [Challenging] • Target: Remainder & Factor Theorems • Cross-Lesson: 1.02 Rational Functions & Asymptotes

For the polynomial $f(x) = 2x^3 + 3x - 1$, describe the end behavior of f.

a. down on the left, up on the right (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

b. up on both ends (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

c. up on the left, down on the right (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

d. down on both ends (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

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40 [Challenging] • Target: Rational Root Theorem & Complex Conjugates • Cross-Lesson: 1.03 Rates of Change & Concavity

For the polynomial $f(x) = 3x^5 + 3x - 1$, describe the end behavior of f.

a. down on the left, up on the right (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

b. down on both ends (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

c. up on both ends (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

d. up on the left, down on the right (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

41 [Challenging] • Target: Leading Term Test and End Behavior • Cross-Lesson: 2.01 Exponential Functions & Growth Models

For the polynomial $f(x) = 2x^5 + 3x - 1$, describe the end behavior of f.

a. up on the left, down on the right (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

b. down on both ends (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

c. up on both ends (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

d. down on the left, up on the right (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

42 [Challenging] • Target: Multiplicity of Zeros and Graph Geometry • Cross-Lesson: 2.02 Logarithmic Functions & Properties

For the polynomial $f(x) = 1x^5 + 3x - 1$, describe the end behavior of f.

a. up on the left, down on the right (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

b. down on both ends (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

c. up on both ends (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

d. down on the left, up on the right (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

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43 [Challenging] • Target: Remainder & Factor Theorems • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

For the polynomial $f(x) = -1x^2 + 2x - 1$, describe the end behavior of f.

a. up on the left, down on the right (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

b. up on both ends (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

c. down on both ends (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

d. down on the left, up on the right (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

44 [Challenging] • Target: Rational Root Theorem & Complex Conjugates • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

For the polynomial $f(x) = -3x^2 + 2x - 1$, describe the end behavior of f.

a. up on the left, down on the right (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

b. up on both ends (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

c. down on the left, up on the right (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

d. down on both ends (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

45 [Challenging] • Target: Leading Term Test and End Behavior • Cross-Lesson: 4.01 Inverse & Composite Functions

For the polynomial $f(x) = -2x^2 + 2x - 1$, describe the end behavior of f.

a. down on the left, up on the right (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

b. down on both ends (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

c. up on both ends (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

d. up on the left, down on the right (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

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